



Title:

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Date: / /

M T W T F S S

$$q1. \overset{7}{1} \overset{6}{0} \overset{5}{1} \overset{4}{0} \overset{3}{0} \overset{2}{1} \overset{1}{0} \times \overset{7}{1} \overset{6}{0} \overset{5}{1} \overset{4}{0} \overset{3}{0} \overset{2}{1} \overset{1}{0} =$$

convert

$$(128 + 32 + 4 + 1) = 165 \times 2 \rightarrow 330_{10}$$

Since it wraps we have to modulus or subtract target from the total range

$$256 | 330 \times r.74$$

$$330 - 256 = 74$$

$$q2. \begin{array}{r} 10111000 \\ - 01000111 \\ \hline \end{array}$$

$$\begin{array}{r} 10001000 \\ - 10001000 \\ \hline \end{array}$$

$$64 + 8 \rightarrow -72_{10}$$

$$2 | -85 \rightarrow 1$$

$$42 \rightarrow 0$$

$$21 \rightarrow 1$$

$$10 \rightarrow 0$$

$$5 \rightarrow 1$$

$$2 \rightarrow 0$$

$$1 \rightarrow 1$$

$$0 \rightarrow$$

$$10101011$$

$$q4. \overset{7}{0} \overset{6}{1} \overset{5}{1} \overset{4}{1} \overset{3}{1} \overset{2}{0} \overset{1}{1} \overset{0}{1} =$$

$$1 + 4 + 2 + 0.5 + 0.125 + 0.0625 = 7.6875$$

b)

$$\overset{4}{1} \overset{3}{0} \overset{2}{1} \overset{1}{0} \overset{0}{1} \overset{-3}{1} \overset{-1}{0} \overset{-1}{1}$$

$$16 + 4 + 1 + 0.5 + 0.125 = 21.625$$

$$q3. \text{Convert } (1023)_3 \rightarrow \text{base } 8$$

$$1 \times 3^3 + 2 \times 3^2 + 3 \times 3^1 = 138_{10}$$

$$8 | 138 \rightarrow 25 \times 8 = 2 \quad 218$$

$$17 \rightarrow 125 \times 8 = 1$$

$$2 \rightarrow 25 \times 8 = 2$$

$$(762)_8 \rightarrow \text{base } 5$$

$$7 \times 8^2 + 6 \times 8^1 + 2 \times 8^0 = 498_{10}$$

$$5 | 498 \rightarrow 19 \times 5 = 3 \quad 343_5$$

$$99 \rightarrow 8 \times 5 = 4$$

$$19 \rightarrow 8 \times 5 = 4$$

$$3 \rightarrow 0.6 \rightarrow 3$$

$$0 \rightarrow$$

$$q5. \begin{array}{c} \text{Sign} \\ 1 \end{array} \begin{array}{c} \text{exponent} \\ 1100 \end{array} \begin{array}{c} \text{mantissa} \\ 101 \end{array}$$

$$x = (-1)^s \times M \times 2^E$$

$$E = \text{Exponent} - \text{bias}$$

$$\text{normalise: } 1.xxxx \times 2^{\text{mantissa}}$$

$$\text{bias} = 12 - 7 = 5$$

$$1.101 \times 2^{12-5}$$

$$\downarrow$$

$$110100$$

$$32 + 16 + 4 + 82 \approx -52$$

$$0 \overset{e}{01010} \overset{e}{010}$$

$$6 - 7 = -2$$

$$e | 010 \times 2^{-2}$$

$$.01010 \rightarrow$$

$$\overset{-1}{0} \overset{-1}{0} \overset{-1}{1} \overset{-1}{0} \overset{-1}{1} \overset{-1}{0}$$

$$0.25 + 0.0625 = 0.3125$$